

Japan all had regimes that encouraged a warrior spirit and praised war as a beneficent furnace in which men would be tempered to hardest steel. Their airmen were expected to triumph as the embodiment of the martial virtues – physical courage, ruthless aggression, patriotic self-sacrifice. Attitudes in the United States, Britain, and even the Soviet Union were more practical and pragmatic. The Soviets were second to none in the sacrifices they demanded of their people, but like the other Allied leaders Stalin knew the war would be won more by economic organization than by martial spirit. Victory in the air required great courage and skill from aircrews, untiring support from ground crews, the work of engineers constructing airfields, and the inventiveness of scientists and aircraft designers. But in the end the air war was won by industrial output.

Levels of productivity

The gulf in productivity between the major combatants was ultimately overwhelming. Japan and Italy simply did not have the industrial capacity or sophistication to keep up as the demands of aerial warfare intensified. The Germans in principle had both the factories and the expertise, but poor use was made of these assets. Disorganization, failure to commit resources, and bad decision-making meant that Germany only moved into top gear in 1943–44, by which time it was being asked to perform miracles under intensive air bombardment and with severe shortages of essential materials.

On the other side, Britain's air industry performed remarkably well – its ability to recover from the losses of front-line aircraft was one of the keys to the country's survival in the Battle of Britain. The achievements of the Soviet wartime aircraft industry would have been outstanding under any circumstances, but were doubly so given the conditions that prevailed after the German invasion of June 1941. Despite the primitive conditions and a lack of skilled personnel, raw materials, and machine tools, the Soviets turned out increasingly effective aircraft in remarkable quantities.

No one, however, could match the United States in mass production. The story of the US air industry in World War II is one of expansion at breakneck speed conducted with, on the whole, astonishing efficiency. The Douglas company offers a good example of the scale of output achieved. Where it had built fewer than 1,000 DC-3s up to 1941, during the war it made some 10,000 C-47s – the military transport

Junkers Ju 52/3m



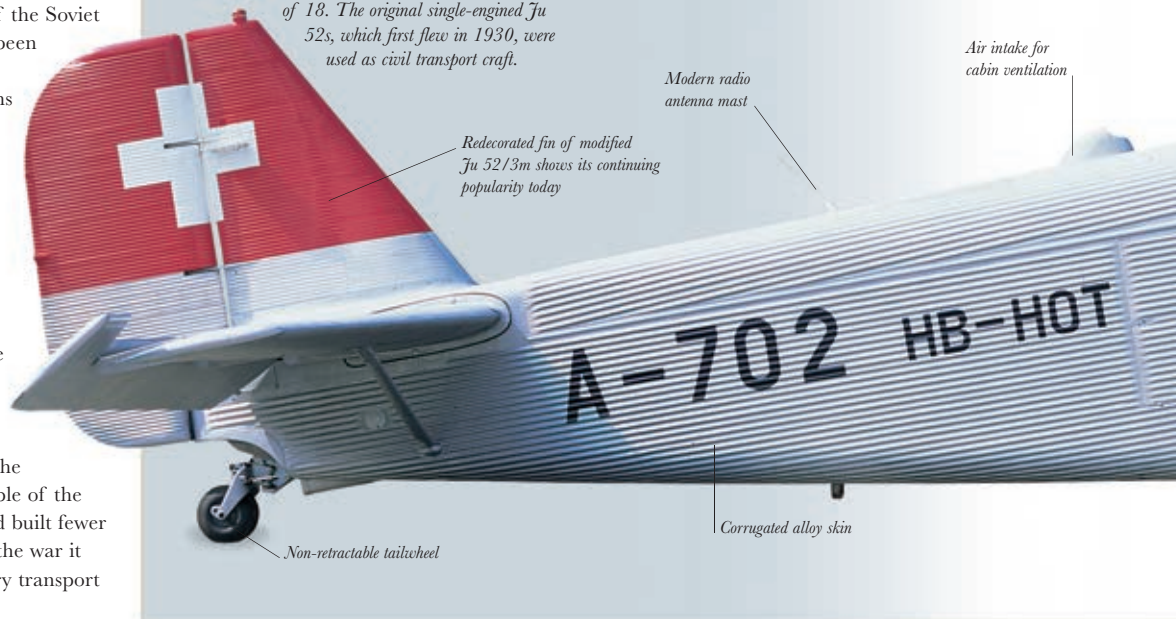
NICKNAMED “TANTE JU” (“Auntie Junkers”), the Ju 52/3m was an aircraft that inspired a special affection in those who flew them. With its corrugated alloy skin and old-fashioned tri-motor configuration, the airplane won no prizes for elegance. Furthermore, it was deafeningly noisy to fly in and desperately slow – the Douglas C-47 could easily outpace it. Yet over 4,800 Ju 52/3m aircraft were produced, outnumbering any other European transport model.

Originally designed as a passenger airliner, the Ju 52/3m was a versatile, rugged, and reliable workhorse, whose main role was carrying troops and supplies. At the start of the Spanish Civil War, they were used to airlift Franco's troops from Morocco into Spain. During World War II, it was the Luftwaffe's major transport aircraft, operating in all weather conditions and over the most inhospitable terrain, from the snowbound Russian steppes to the Tunisian desert, from the Norwegian fjords to the mountains of Crete. In May 1941, nearly 500 were used to fly German paratroopers from mainland Greece and drop them over Crete in a spectacular aerial assault. In the terrible winter of 1942–43, they flew supplies into the frozen airstrips around Stalingrad and evacuated the wounded. As their world fell apart during the final years of the war, German servicemen took comfort from flying in an aircraft that had become so familiar and never seemed to let them down.



NARROW CORRIDOR

The Ju 52/3m's long, narrow cabin, with a single row of seats on each side, had a maximum passenger capacity of 18. The original single-engined Ju 52s, which first flew in 1930, were used as civil transport craft.



Redecorated fin of modified Ju 52/3m shows its continuing popularity today

Modern radio antenna mast

Air intake for cabin ventilation

Corrugated alloy skin

Non-retractable tailwheel